

nanoRMS：同时定位不同RNA修饰

安装

```
wget https://github.com/novoalab/nanoRMS/archive/refs/tags/v1.0.tar.gz && tar -zxvf v1.0.tar.gz
```

环境配置

###docker build -f /home/chen/jiangjing/soft/nanoRMS-1.0/docker/Dockerfile . （不可用，内置有问题）

###linux配置所需依赖包

```
conda create nanoRMS
```

```
conda activate nanoRMS
```

```
conda install bioconda/label/cf201901::ont-tombo hdf5 minimap2 dask nanopolish
```

```
pip install "numpy<1.20" ont_fast5_api matplotlib mappy numba pysam pandas seaborn  
scikit-learn jupyterlab openpyxl mappy scipy
```

###linux R配置包

###环境：当前环境R版本低，需安装版本大于3.5，安装依赖包

①conda uninstall r-base

②conda install conda-forge::r-base

③install.packages(c("optparse","stringr","dplyr","plyr","VennDiagram","RColorBrewer","data.table"))

使用

1. prediction of RNA modified sits

1.1 Extract base-calling features using Epinano-RMS

① 对reference fasta构建索引，并且同一目录下需要有fai文件（samtools faidx）

```
java -jar epinano_RMS/picard.jar CreateSequenceDictionary  
REFERENCE=reference.fasta OUTPUT= reference.fasta.dict
```

脚本示例：java -jar /home/chen/jiangjing/soft/nanoRMS/epinano_RMS/picard.jar
CreateSequenceDictionary REFERENCE=yeast_all_rRNA.fa OUTPUT= reference.fasta.dict

② 计算碱基位点频率

```
python3 epinano_RMS/epinano_rms.py -R <reference_file> -b <bam_file> -s  
epinano_RMS/sam2tsv
```

脚本: python3 /home/chen/jiangjing/soft/nanoRMS/epinano_RMS/epinano_rms.py -R /home/chen/jiangjing/soft/nanoRMS/predict_rna_mod/references/yeast_all_rRNA.fa -b /home/chen/jiangjing/soft/nanoRMS/predict_rna_mod/test_data/test.bam -s /home/chen/jiangjing/soft/nanoRMS/epinano_RMS/sam2tsv.jar

提醒会报错, 显示无某个文件夹, 直接建一个, 后面删掉就行: mkdir -p ./home/chen/jiangjing/soft/nanoRMS/predict_rna_mod/test_data/test.tmp_splitting_base_freq.done_splitting

结果文件: 包含每个碱基位点频率的.csv文件 (test.per.site.baseFreq.csv)

```
~[0] <> head -n20 test.per.site.baseFreq.csv
#Ref,pos,base,strand,cov,mean_q,median_q,std_q,mis,ins,del,ACGT
18s,1,T,+,2,11.500000,11.500000,4.500000,0.000000,0.000000,0.000000,0:0:0:2
18s,10,G,+,2,22.500000,22.500000,3.500000,0.000000,0.000000,0.000000,0:0:2:0
18s,100,A,+,17,12.000000,10.000000,8.538682,0.058824,0.117647,0.352941,10:0:1:0
18s,1000,C,+,37,10.675676,9.000000,6.169034,0.000000,0.189189,0.000000,0:37:0:0
18s,1001,A,+,37,13.108108,13.000000,7.043432,0.027027,0.405405,0.000000,36:0:0:1
18s,1002,G,+,37,24.756757,26.000000,8.112071,0.000000,0.243243,0.000000,0:0:37:0
18s,1003,A,+,37,24.351351,25.000000,7.121404,0.000000,0.081081,0.000000,37:0:0:0
18s,1004,T,+,37,22.189189,25.000000,7.479116,0.000000,0.054054,0.000000,0:0:0:37
```

1.2 Predict RNA modifications

① Single sample RNA modification prediction (i.e. "de novo" prediction)

##predicting pseudouridine RNA modifications in mitochondrial rRNAs, validating 2 out of the 2 sites that were predicted in all 3 biological replicates. Specifically, **pseudouridine causes strong mismatch signatures** in the modified position, largely **in the form of C-to-U mismatches**

```
Rscript --vanilla Pseudou_prediction_singlecondition.R [options] -f
<epinano_file1> (-s <epinano_file2> -t <epinano_file3>)
```

Options:

```
-h, --help
    Show this help message and exit

-f CHARACTER, --epinano1=CHARACTER
    first epinano file

-s CHARACTER, --epinano2=CHARACTER
    second epinano file

-t CHARACTER, --epinano3=CHARACTER
    third epinano file

-p CHARACTER, --modpos=CHARACTER
    mod positions file [default=
positions/RNA_Mod_Positions_rRNAyeast.tsv]

-m NUMBER, --misfreq=NUMBER
    Mismatch frequency threshold [default= 0.137]

-c NUMBER, --Cfreq=NUMBER
    C mismatch frequency threshold [default= 0.578]
```

```
脚本示例: Rscript --vanilla
/home/chen/jiangjing/soft/nanoRMS/predict_rna_mod/Pseudou_prediction_singlecondition.R -f
WT_rRNA_Epinano.csv -s sn34KO_rRNA_Epinano.csv -t sn36KO_rRNA_Epinano.csv -p
../positions/RNA_Mod_Positions_rRNA_Yeast.tsv
```

##csv文件是1.1的结果文件

##其中mod_position文件是依据已知的假尿嘧啶修饰位点相关数据库和文献, 自建的

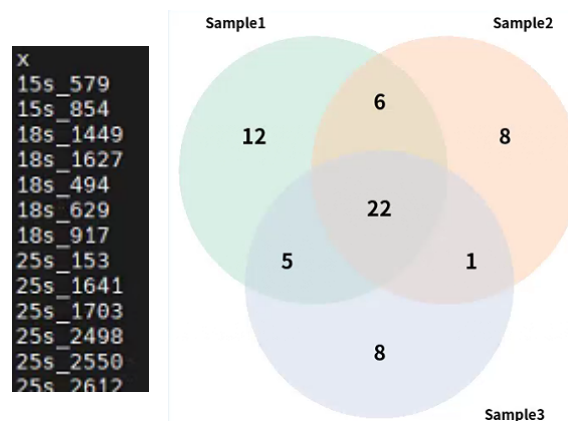
##mod_position文件相关解释: <https://github.com/novoalab/nanoRMS/issues/32>

##找到的修饰位点相关数据库: <https://rna.sysu.edu.cn/rmbase3/pseudosearch.php>

结果文件: (1) threesample_predicted_reproducible_y_sites.tsv (三个重复样本交集位点)

(2) 三个重复样本交集venn图:

Threesamples_venn_diagramm_mitochondrial.tiff



① Paired sample RNA modification prediction (i.e. "differential-error"-based prediction)

##Pseudouridine is not always present in high stoichiometries (e.g. rRNAs), but can also be present in low stoichiometries (e.g. in mRNAs). Please note that pseudouridines in mRNAs cannot be accurately predicted using "de novo" mode, because the background nanopore 'error' is too similar to the 'error' caused by the presence of pseudouridine. For such cases, we can predict differentially pseudouridylated sites by identifying which sites show pseudouridine differential error signatures between two conditions, as shown below. This type of pairwise comparison can be done for WT-KO, or between two conditions (e.g. normal-heat stress). In the image below, some examples of heat-responsive sites that were identified using this script are shown. Differential-error based prediction can be applied to any type of RNA (mRNA, snoRNAs, snRNAs, rRNAs etc).

1) For Transcriptome mapped reads (Reads from one strand)

```
Rscript --vanilla Pseudou_prediction_pairedcondition_transcript.R [options] -f
<epinano_file1> -s <epinano_file2>
```

Options:

```
-h, --help
    Show this help message and exit

-f CHARACTER, --epinano1=CHARACTER
    first epinano file
```

```

-s CHARACTER, --epinano2=CHARACTER
    second epinano file

-p CHARACTER, --modpos=CHARACTER
    mod positions file
    [default=positions/RNA_Mod_Positions_ncRNAyeast_HeatSensitive.tsv]

-m NUMBER, --misfreq=NUMBER
    Mismatch frequency threshold [default= 0.137]

-c NUMBER, --Cfreq=NUMBER
    C mismatch frequency threshold [default= 0.578]

-d NUMBER, --diff=NUMBER
    mismatch frequency difference threshold [default= 0.1]

```

脚本示例: Rscript --vanilla ../Pseudou_prediction_pairedcondition_transcript.R -f WT_ncRNA_Normal_Rep1_Epinano.csv -s WT_ncRNA_HeatShock_Rep1_Epinano.csv -p ../positions/RNA_Mod_Positions_ncRNAyeast_HeatSensitive.tsv

结果文件: (1) Paired_comparison_altering_sites_pU_predictions.bed

(2) Paired_comparison_altering_sites_pU_predictions.tsv

```

==> Paired_comparison_altering_sites_pU_predictions.bed <==
LSR1_snRNA 453 454 . 1 +
LSR1_snRNA 627 628 . 1 +
RDN18-2_rRNA 1143 1144 . 1 +
RDN18-2_rRNA 1185 1186 . 1 +
RDN18-2_rRNA 1186 1187 . 1 +
RDN18-2_rRNA 1289 1290 . 1 +
RDN18-2_rRNA 1626 1627 . 1 +
RDN18-2_rRNA 1769 1770 . 1 +
RDN37-2_rRNA 1994 1995 . 1 +
RPR1_ncRNA 329 330 . 1 +

==> Paired_comparison_altering_sites_pU_predictions.tsv <==
Chr Pos Chr_Pos Base Pseudouridine Mis.data1 C_freq.data1 Mis.data2 C_freq.data2 Mis.difference
LSR1_snRNA 454 LSR1_snRNA_454 T Yes(HeatSensitive) 0.08125 0.333333333333333 0.20896 0.678571428571429 -0.12771
LSR1_snRNA 628 LSR1_snRNA_628 T Yes(HeatSensitive) 0.02414 0.75 0.21831 0.935483870967742 -0.19417
RDN18-2_rRNA 1144 RDN18-2_rRNA_1144 T No 0.17143 0.833333333333333 0.05085 0.666666666666667 0.12058
RDN18-2_rRNA 1186 RDN18-2_rRNA_1186 T No 0.5 0.722222222222222 0.25 0.8 0.25
RDN18-2_rRNA 1187 RDN18-2_rRNA_1187 T No 0.25 0.666666666666667 0.11667 0.857142857142857 0.13333
RDN18-2_rRNA 1290 RDN18-2_rRNA_1290 T No 0.58537 0.875 0.39394 0.884615384615385 0.19143
RDN18-2_rRNA 1627 RDN18-2_rRNA_1627 T No 0.19149 0.777777777777778 0.06667 1 0.12482
RDN18-2_rRNA 1770 RDN18-2_rRNA_1770 T No 0.17021 0.25 0.27027 0.6 -0.10006
RDN37-2_rRNA 1995 RDN37-2_rRNA_1995 T No 0.28453 0.592314901593252 0.13851 0.56695464362851 0.14602

```

2) For Genome mapped reads (Reads from both strands)

① csv转bed

```
./Epinano_to_BED.sh <epinano_file1>
```

脚本示例: ./Epinano_to_BED.sh WT_mRNA_Normal_Epinano.csv

② Convert GTF (Only CDS) output into BED

```
Rscript --vanilla GTF_to_BED.R <GTF_File>
```

脚本示例: Rscript --vanilla ../GTF_to_BED.R Saccr64.gtf

③ Intersect Epinano_BED file and GTF_BED file

```
bedtools intersect -a <Epinano_Bed> -b <GTF_Bed> -wa -wb > output.bed
```

脚本示例: bedtools intersect -a ../test_data/WT_mRNA_Normal_Epinano.csv.bed -b Saccar3.bed -wa -wb > WT_mRNA_Normal_Epinano_final.bed

④mapped

官网给的是错的:

~~Rscript --vanilla ../Pseudou_prediction_pairedcondition_genome.R -f WT_mRNA_Epinano.csv -s sn36KO_rRNA_Epinano.csv~~

第一步: 修改脚本: Pseudou_prediction_pairedcondition_genome.R

```
##FUNCTION DEFINED
processing <- function(data) {
  #Import the mod file
  mod.file <- read.delim(opt$modpos, sep="") #opt$modpos Mod positions file RNA_Mod_Positions_mRNAyeast_HeatSensitive.tsv
```

mod.file去掉#

第二步: 脚本示例:

Rscript --vanilla ../Pseudou_prediction_pairedcondition_genome.R -f WT_mRNA_Normal_Epinano_final.bed -s WT_mRNA_HeatShock_Epinano_final.bed -p ../positions/RNA_Mod_Positions_mRNAyeast_HeatSensitive.tsv

结果文件: (1) Paired_comparison_altering_sites_pU_predictions.bed

(2) Paired_comparison_altering_sites_pU_predictions.tsv

```
==== Paired comparison altering_sites_pU_predictions.bed ====
chrI 131179 131180 . 1 +
chrI 131304 131305 . 1 +
chrI 57958 57959 . 1 -
chrI 58406 58407 . 1 -
chrII 257240 257241 . 1 -
chrII 306545 306546 . 1 -
chrII 338839 338840 . 1 -
chrII 338990 338991 . 1 -
chrII 372709 372710 . 1 -
chrII 372713 372714 . 1 -

==== Paired comparison altering_sites_pU_predictions.tsv ====
chr Pos Chr_Pos_Base Pseudouridine Strand.data1 cds_chr.data1 cds_start.data1 cds_end.data1 cds_strand.data1 cds_name.data1 Coverage.data1 Mis.data1 C_freq.data1 Gene_N
ane.data1 Gene_Pos.data1 Mis.difference cds_chr.data2 cds_start.data2 cds_end.data2 cds_strand.data2 Coverage.data2 Mis.data2 C_freq.data2 Gene_Name.data2 Gene_P
os.data2
chrI 131180 chrI_131180 T No + chrI 130799 131980 + YAL012W 66 0.0303 1 No No + chrI 130799 131980 + YAL012W 103 0.15534 0.9375
No No -0.12504
chrI 131305 chrI_131305 T No + chrI 130799 131980 + YAL012W 75 0.24 0.9444444444444444 No No + chrI 130799 131980 + YAL012W 113 0
.35398 0.95 No No -0.11398
chrI 57959 chrI_57959 A Yes - chrI 57953 58462 - YAL044C 146 0.06164 0.8888888888888889 YAL044C 504 - chrI 57953 58462 - YAL044C 128 0
.20312 1 YAL044C 504 -0.14148
chrI 58407 chrI_58407 A No - chrI 57953 58462 - YAL044C 99 0.20202 0.85 No No - chrI 57953 58462 - YAL044C 94 0.05319 0.8 N
0 No 0.14883
chrII 257241 chrII_257241 A No - chrII 257115 257975 - YBR011C 36 0.22222 1 No No - chrII 257115 257975 - YBR011C 107 0.11215 1 N
0 No 0.11007
chrII 306546 chrII_306546 A No - chrII 306272 306955 - YBR035C 37 0.08108 1 No No - chrII 306272 306955 - YBR035C 31 0.22581 0.8571
42857142857 No No -0.14473
```

2. RNA modification stoichiometry estimation using Nanopolish resquigling (官网不推荐)

3. RNA modification stoichiometry estimation using Tombo resquigling

3.1 第一步下载相关数据的fast5及fq文件

3.2 retrieve per-read features from all sample (fast5转bam)

```
./get_features.py --rna -f reference.fa -t 6 -i .fast5文件路径
```

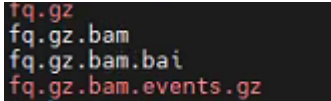
结果文件:

```
batch0.fast5
batch0.fast5.bam
batch0.fast5.bam.bai
batch0.fast5.bam.json
```

3.3 Run nanopolish if you wish to compare it with tombo. You can skip this step if you don't want to compare the tools.

```
ref=reference.fa
for d in guppy3.0.3.hac/*_{snR*,wt}/; do
  if [ ! -s $d/fq.gz.bam.events.gz ]; then
    echo `date` $d;
    cat $d/*.fastq.gz > $d/fq.gz;
    nanopolish index -d $d $d/fq.gz 2> /dev/null;
    minimap2 -ax map-ont $ref $d/fq.gz 2> /dev/null | samtools sort -o
    $d/fq.gz.bam;
    samtools index $d/fq.gz.bam;
    nanopolish eventalign -n -t 6 -q 10 --progress --signal-index --scale-events -
    -reads $d/fq.gz --bam $d/fq.gz.bam --genome $ref | gzip >
    $d/fq.gz.bam.events.gz;
  fi;
done; date
```

结果文件:



3.4 Estimate modification frequency difference between two samples

####Note, you'll need to provide candidate positions that are likely modified. Those were identified earlier -- please see above section 1.2. Predict RNA modifications. so here we'll just generate BED file from existing candidate file.

```
# prepare BED - this is no longer needed step!
f=per_read/results/predictions_ncRNA_WT30C_WT45C.tsv.gz
zgrep -v X.Ref $f |awk -F'\t' 'BEGIN {OFS = FS} {print $1,$2-1,$2,".",100,"+"}'
> $f.bed
```

bed格式:

snR86_snoRNA	395	396	.	100	+
snR86_snoRNA	455	456	.	100	+
snR86_snoRNA	456	457	.	100	+
snR86_snoRNA	530	531	.	100	+
snR86_snoRNA	536	537	.	100	+
snR86_snoRNA	559	560	.	100	+
LSR1_snoRNA	627	628	.	100	+
snR37_snoRNA	204	205	.	100	+
snR37_snoRNA	210	211	.	100	+
snR86_snoRNA	458	459	.	100	+
snR17a_snoRNA	211	212	.	100	+
snR37_snoRNA	115	116	.	100	+

```
# calculate modification frequency difference between control and sample of
interest
per_read/get_freq.py -f $ref -b $f.bed -o $f.bed.tsv.gz -1
per_read/guppy3.0.3.hac/*WT30C/workspace/*.fast5.bam -2
per_read/guppy3.0.3.hac/*WT45C/workspace/*.fast5.bam
```

usage: get_freq.py

option:

```
-h/--help
Show this help message and exit
```



```

--version/-v

-1 control...

-2 sample...

-o output

-b BED文件

-f/--fasta

-m mincov (required minimum number of reads,but it has to be at least 5 due to
KNN requirements.)

```

结果文件: predictions_ncRNA_WT30C_WT45C.bed.tsv.gz

(KNN 的最终目的是分类, 而 K-means 的目的是给所有距离相近的点分配一个类别, 也就是聚类)

chrom	pos	strand	min coverage	mod_freq	diff_kmeans	mod_freq	diff_knn
snR86	snoRNA	396	+	149	0.10761735183047333	0.43426536482282274	
snR86	snoRNA	456	+	149	-0.06959907644745333	0.7271997965053711	
snR86	snoRNA	457	+	149	0.6046921165398086	0.6493044005713503	
snR86	snoRNA	531	+	149	0.08754182401627958	0.36742520594047784	
snR86	snoRNA	537	+	149	0.6934275148218444	0.6954037607372767	
snR86	snoRNA	560	+	149	-0.44714031345999566	0.6580507562564815	
LSR1	snoRNA	628	+	278	0.08354108792174181	0.3203694470166441	
snR37	snoRNA	205	+	1858	0.3519216405614991	0.6108988159311087	
snR37	snoRNA	211	+	1858	0.5405702862415149	0.6337119131829265	
snR86	snoRNA	459	+	149	-0.6047508169135344	0.6551353043614377	
snR17a	snoRNA	212	+	324	0.5894798157109572	0.5754393733329116	
snR37	snoRNA	116	+	1857	0.02025265679777566	0.44149275218972883	

4. Visualization of per-read current intensities at individual sites

4.1 Data pre-processing

Firstly, generate a collapsed Nanopolish event align output(3.3结果文件), by collapsing all the multiple observations for a given position from a same read.

```
python3 per_read_mean.py <event_align_file>
```

脚本示例: python3 per_read_mean.py test_data/data1_eventalign_output.txt

输出文件: data1_eventalign_output.txt_processed_perpos_mean.csv

```

cat data1_eventalign_output.txt | \
awk 'BEGIN{FS=" "}{print $1,$2,$3,$4,$5,$6,$7,$8,$9,$10,$11,$12,$13,$14,$15,$16,$17,$18,$19,$20,$21,$22,$23,$24,$25,$26,$27,$28,$29,$30,$31,$32,$33,$34,$35,$36,$37,$38,$39,$40,$41,$42,$43,$44,$45,$46,$47,$48,$49,$50,$51,$52,$53,$54,$55,$56,$57,$58,$59,$60,$61,$62,$63,$64,$65,$66,$67,$68,$69,$70,$71,$72,$73,$74,$75,$76,$77,$78,$79,$80,$81,$82,$83,$84,$85,$86,$87,$88,$89,$90,$91,$92,$93,$94,$95,$96,$97,$98,$99,$100,$101,$102,$103,$104,$105,$106,$107,$108,$109,$110,$111,$112,$113,$114,$115,$116,$117,$118,$119,$120,$121,$122,$123,$124,$125,$126,$127,$128,$129,$130,$131,$132,$133,$134,$135,$136,$137,$138,$139,$140,$141,$142,$143,$144,$145,$146,$147,$148,$149,$150,$151,$152,$153,$154,$155,$156,$157,$158,$159,$160,$161,$162,$163,$164,$165,$166,$167,$168,$169,$170,$171,$172,$173,$174,$175,$176,$177,$178,$179,$180,$181,$182,$183,$184,$185,$186,$187,$188,$189,$190,$191,$192,$193,$194,$195,$196,$197,$198,$199,$200,$201,$202,$203,$204,$205,$206,$207,$208,$209,$210,$211,$212,$213,$214,$215,$216,$217,$218,$219,$220,$221,$222,$223,$224,$225,$226,$227,$228,$229,$230,$231,$232,$233,$234,$235,$236,$237,$238,$239,$240,$241,$242,$243,$244,$245,$246,$247,$248,$249,$250,$251,$252,$253,$254,$255,$256,$257,$258,$259,$260,$261,$262,$263,$264,$265,$266,$267,$268,$269,$270,$271,$272,$273,$274,$275,$276,$277,$278,$279,$280,$281,$282,$283,$284,$285,$286,$287,$288,$289,$290,$291,$292,$293,$294,$295,$296,$297,$298,$299,$300,$301,$302,$303,$304,$305,$306,$307,$308,$309,$310,$311,$312,$313,$314,$315,$316,$317,$318,$319,$320,$321,$322,$323,$324,$325,$326,$327,$328,$329,$330,$331,$332,$333,$334,$335,$336,$337,$338,$339,$340,$341,$342,$343,$344,$345,$346,$347,$348,$349,$350,$351,$352,$353,$354,$355,$356,$357,$358,$359,$360,$361,$362,$363,$364,$365,$366,$367,$368,$369,$370,$371,$372,$373,$374,$375,$376,$377,$378,$379,$380,$381,$382,$383,$384,$385,$386,$387,$388,$389,$390,$391,$392,$393,$394,$395,$396,$397,$398,$399,$400,$401,$402,$403,$404,$405,$406,$407,$408,$409,$410,$411,$412,$413,$414,$415,$416,$417,$418,$419,$420,$421,$422,$423,$424,$425,$426,$427,$428,$429,$430,$431,$432,$433,$434,$435,$436,$437,$438,$439,$440,$441,$442,$443,$444,$445,$446,$447,$448,$449,$450,$451,$452,$453,$454,$455,$456,$457,$458,$459,$460,$461,$462,$463,$464,$465,$466,$467,$468,$469,$470,$471,$472,$473,$474,$475,$476,$477,$478,$479,$480,$481,$482,$483,$484,$485,$486,$487,$488,$489,$490,$491,$492,$493,$494,$495,$496,$497,$498,$499,$500,$501,$502,$503,$504,$505,$506,$507,$508,$509,$510,$511,$512,$513,$514,$515,$516,$517,$518,$519,$520,$521,$522,$523,$524,$525,$526,$527,$528,$529,$530,$531,$532,$533,$534,$535,$536,$537,$538,$539,$540,$541,$542,$543,$544,$545,$546,$547,$548,$549,$550,$551,$552,$553,$554,$555,$556,$557,$558,$559,$560,$561,$562,$563,$564,$565,$566,$567,$568,$569,$570,$571,$572,$573,$574,$575,$576,$577,$578,$579,$580,$581,$582,$583,$584,$585,$586,$587,$588,$589,$590,$591,$592,$593,$594,$595,$596,$597,$598,$599,$600,$601,$602,$603,$604,$605,$606,$607,$608,$609,$610,$611,$612,$613,$614,$615,$616,$617,$618,$619,$620,$621,$622,$623,$624,$625,$626,$627,$628,$629,$630,$631,$632,$633,$634,$635,$636,$637,$638,$639,$640,$641,$642,$643,$644,$645,$646,$647,$648,$649,$650,$651,$652,$653,$654,$655,$656,$657,$658,$659,$660,$661,$662,$663,$664,$665,$666,$667,$668,$669,$670,$671,$672,$673,$674,$675,$676,$677,$678,$679,$680,$681,$682,$683,$684,$685,$686,$687,$688,$689,$690,$691,$692,$693,$694,$695,$696,$697,$698,$699,$700,$701,$702,$703,$704,$705,$706,$707,$708,$709,$710,$711,$712,$713,$714,$715,$716,$717,$718,$719,$720,$721,$722,$723,$724,$725,$726,$727,$728,$729,$730,$731,$732,$733,$734,$735,$736,$737,$738,$739,$740,$741,$742,$743,$744,$745,$746,$747,$748,$749,$750,$751,$752,$753,$754,$755,$756,$757,$758,$759,$760,$761,$762,$763,$764,$765,$766,$767,$768,$769,$770,$771,$772,$773,$774,$775,$776,$777,$778,$779,$780,$781,$782,$783,$784,$785,$786,$787,$788,$789,$790,$791,$792,$793,$794,$795,$796,$797,$798,$799,$800,$801,$802,$803,$804,$805,$806,$807,$808,$809,$810,$811,$812,$813,$814,$815,$816,$817,$818,$819,$820,$821,$822,$823,$824,$825,$826,$827,$828,$829,$830,$831,$832,$833,$834,$835,$836,$837,$838,$839,$840,$841,$842,$843,$844,$845,$846,$847,$848,$849,$850,$851,$852,$853,$854,$855,$856,$857,$858,$859,$860,$861,$862,$863,$864,$865,$866,$867,$868,$869,$870,$871,$872,$873,$874,$875,$876,$877,$878,$879,$880,$881,$882,$883,$884,$885,$886,$887,$888,$889,$890,$891,$892,$893,$894,$895,$896,$897,$898,$899,$900,$901,$902,$903,$904,$905,$906,$907,$908,$909,$910,$911,$912,$913,$914,$915,$916,$917,$918,$919,$920,$921,$922,$923,$924,$925,$926,$927,$928,$929,$930,$931,$932,$933,$934,$935,$936,$937,$938,$939,$940,$941,$942,$943,$944,$945,$946,$947,$948,$949,$950,$951,$952,$953,$954,$955,$956,$957,$958,$959,$960,$961,$962,$963,$964,$965,$966,$967,$968,$969,$970,$971,$972,$973,$974,$975,$976,$977,$978,$979,$980,$981,$982,$983,$984,$985,$986,$987,$988,$989,$990,$991,$992,$993,$994,$995,$996,$997,$998,$999,$1000,$1001,$1002,$1003,$1004,$1005,$1006,$1007,$1008,$1009,$1010,$1011,$1012,$1013,$1014,$1015,$1016,$1017,$1018,$1019,$1020,$1021,$1022,$1023,$1024,$1025,$1026,$1027,$1028,$1029,$1030,$1031,$1032,$1033,$1034,$1035,$1036,$1037,$1038,$1039,$1040,$1041,$1042,$1043,$1044,$1045,$1046,$1047,$1048,$1049,$1050,$1051,$1052,$1053,$1054,$1055,$1056,$1057,$1058,$1059,$1060,$1061,$1062,$1063,$1064,$1065,$1066,$1067,$1068,$1069,$1070,$1071,$1072,$1073,$1074,$1075,$1076,$1077,$1078,$1079,$1080,$1081,$1082,$1083,$1084,$1085,$1086,$1087,$1088,$1089,$1090,$1091,$1092,$1093,$1094,$1095,$1096,$1097,$1098,$1099,$1100,$1101,$1102,$1103,$1104,$1105,$1106,$1107,$1108,$1109,$1110,$1111,$1112,$1113,$1114,$1115,$1116,$1117,$1118,$1119,$1120,$1121,$1122,$1123,$1124,$1125,$1126,$1127,$1128,$1129,$1130,$1131,$1132,$1133,$1134,$1135,$1136,$1137,$1138,$1139,$1140,$1141,$1142,$1143,$1144,$1145,$1146,$1147,$1148,$1149,$1150,$1151,$1152,$1153,$1154,$1155,$1156,$1157,$1158,$1159,$1160,$1161,$1162,$1163,$1164,$1165,$1166,$1167,$1168,$1169,$1170,$1171,$1172,$1173,$1174,$1175,$1176,$1177,$1178,$1179,$1180,$1181,$1182,$1183,$1184,$1185,$1186,$1187,$1188,$1189,$1190,$1191,$1192,$1193,$1194,$1195,$1196,$1197,$1198,$1199,$1200,$1201,$1202,$1203,$1204,$1205,$1206,$1207,$1208,$1209,$1210,$1211,$1212,$1213,$1214,$1215,$1216,$1217,$1218,$1219,$1220,$1221,$1222,$1223,$1224,$1225,$1226,$1227,$1228,$1229,$1230,$1231,$1232,$1233,$1234,$1235,$1236,$1237,$1238,$1239,$1240,$1241,$1242,$1243,$1244,$1245,$1246,$1247,$1248,$1249,$1250,$1251,$1252,$1253,$1254,$1255,$1256,$1257,$1258,$1259,$1260,$1261,$1262,$1263,$1264,$1265,$1266,$1267,$1268,$1269,$1270,$1271,$1272,$1273,$1274,$1275,$1276,$1277,$1278,$1279,$1280,$1281,$1282,$1283,$1284,$1285,$1286,$1287,$1288,$1289,$1290,$1291,$1292,$1293,$1294,$1295,$1296,$1297,$1298,$1299,$1300,$1301,$1302,$1303,$1304,$1305,$1306,$1307,$1308,$1309,$1310,$1311,$1312,$1313,$1314,$1315,$1316,$1317,$1318,$1319,$1320,$1321,$1322,$1323,$1324,$1325,$1326,$1327,$1328,$1329,$1330,$1331,$1332,$1333,$1334,$1335,$1336,$1337,$1338,$1339,$1340,$1341,$1342,$1343,$1344,$1345,$1346,$1347,$1348,$1349,$1350,$1351,$1352,$1353,$1354,$1355,$1356,$1357,$1358,$1359,$1360,$1361,$1362,$1363,$1364,$1365,$1366,$1367,$1368,$1369,$1370,$1371,$1372,$1373,$1374,$1375,$1376,$1377,$1378,$1379,$1380,$1381,$1382,$1383,$1384,$1385,$1386,$1387,$1388,$1389,$1390,$1391,$1392,$1393,$1394,$1395,$1396,$1397,$1398,$1399,$1400,$1401,$1402,$1403,$1404,$1405,$1406,$1407,$1408,$1409,$1410,$1411,$1412,$1413,$1414,$1415,$1416,$1417,$1418,$1419,$1420,$1421,$1422,$1423,$1424,$1425,$1426,$1427,$1428,$1429,$1430,$1431,$1432,$1433,$1434,$1435,$1436,$1437,$1438,$1439,$1440,$1441,$1442,$1443,$1444,$1445,$1446,$1447,$1448,$1449,$1450,$1451,$1452,$1453,$1454,$1455,$1456,$1457,$1458,$1459,$1460,$1461,$1462,$1463,$1464,$1465,$1466,$1467,$1468,$1469,$1470,$1471,$1472,$1473,$1474,$1475,$1476,$1477,$1478,$1479,$1480,$1481,$1482,$1483,$1484,$1485,$1486,$1487,$1488,$1489,$1490,$1491,$1492,$1493,$1494,$1495,$1496,$1497,$1498,$1499,$1500,$1501,$1502,$1503,$1504,$1505,$1506,$1507,$1508,$1509,$1510,$1511,$1512,$1513,$1514,$1515,$1516,$1517,$1518,$1519,$1520,$1521,$1522,$1523,$1524,$1525,$1526,$1527,$1528,$1529,$1530,$1531,$1532,$1533,$1534,$1535,$1536,$1537,$1538,$1539,$1540,$1541,$1542,$1543,$1544,$1545,$1546,$1547,$1548,$1549,$1550,$1551,$1552,$1553,$1554,$1555,$1556,$1557,$1558,$1559,$1560,$1561,$1562,$1563,$1564,$1565,$1566,$1567,$1568,$1569,$1570,$1571,$1572,$1573,$1574,$1575,$1576,$1577,$1578,$1579,$1580,$1581,$1582,$1583,$1584,$1585,$1586,$1587,$1588,$1589,$1590,$1591,$1592,$1593,$1594,$1595,$1596,$1597,$1598,$1599,$1600,$1601,$1602,$1603,$1604,$1605,$1606,$1607,$1608,$1609,$1610,$1611,$1612,$1613,$1614,$1615,$1616,$1617,$1618,$1619,$1620,$1621,$1622,$1623,$1624,$1625,$1626,$1627,$1628,$1629,$1630,$1631,$1632,$1633,$1634,$1635,$1636,$1637,$1638,$1639,$1640,$1641,$1642,$1643,$1644,$1645,$1646,$1647,$1648,$1649,$1650,$1651,$1652,$1653,$1654,$1655,$1656,$1657,$1658,$1659,$1660,$1661,$1662,$1663,$1664,$1665,$1666,$1667,$1668,$1669,$1670,$1671,$1672,$1673,$1674,$1675,$1676,$1677,$1678,$1679,$1680,$1681,$1682,$1683,$1684,$1685,$1686,$1687,$1688,$1689,$1690,$1691,$1692,$1693,$1694,$1695,$1696,$1697,$1698,$1699,$1700,$1701,$1702,$1703,$1704,$1705,$1706,$1707,$1708,$1709,$1710,$1711,$1712,$1713,$1714,$1715,$1716,$1717,$1718,$1719,$1720,$1721,$1722,$1723,$1724,$1725,$1726,$1727,$1728,$1729,$1730,$1731,$1732,$1733,$1734,$1735,$1736,$1737,$1738,$1739,$1740,$1741,$1742,$1743,$1744,$1745,$1746,$1747,$1748,$1749,$1750,$1751,$1752,$1753,$1754,$1755,$1756,$1757,$1758,$1759,$1760,$1761,$1762,$1763,$1764,$1765,$1766,$1767,$1768,$1769,$1770,$1771,$1772,$1773,$1774,$1775,$1776,$1777,$1778,$1779,$1780,$1781,$1782,$1783,$1784,$1785,$1786,$1787,$1788,$1789,$1790,$1791,$1792,$1793,$1794,$1795,$1796,$1797,$1798,$1799,$1800,$1801,$1802,$1803,$1804,$1805,$1806,$1807,$1808,$1809,$1810,$1811,$1812,$1813,$1814,$1815,$1816,$1817,$1818,$1819,$1820,$1821,$1822,$1823,$1824,$1825,$1826,$1827,$1828,$1829,$1830,$1831,$1832,$1833,$1834,$1835,$1836,$1837,$1838,$1839,$1840,$1841,$1842,$1843,$1844,$1845,$1846,$1847,$1848,$1849,$1850,$1851,$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```

```
head -n5 data1_window_file.tsv
contig position reference_kmer read_index event_level_mean Pos sample modification reference
236208 25s 2819 ATAAC 39339 85.84 25s_2819 data1 25s_2826 -7
236209 25s 2819 ATAAC 39401 81.92 25s_2819 data1 25s_2826 -7
236210 25s 2819 ATAAC 39618 86.58 25s_2819 data1 25s_2826 -7
236211 25s 2819 ATAAC 39695 82.07 25s_2819 data1 25s_2826 -7
```

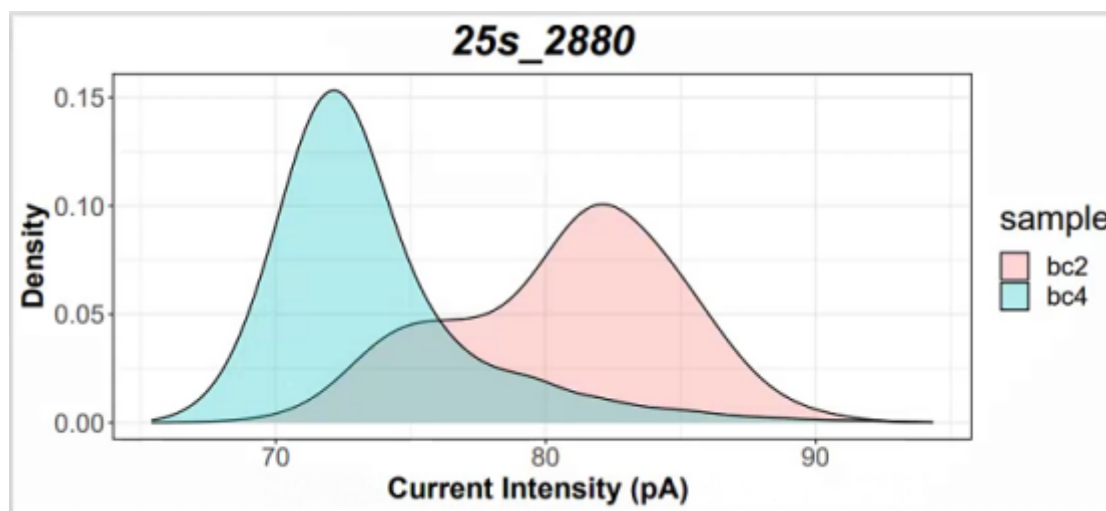
4.2. Visualization of current intensity information:

4.2.1 Density plots

```
Rscript --vanilla density_nanopolish.R <window_file1> <window_file2>
<window_file3(optional)> <window_file4(optional)>
```

脚本示例: Rscript --vanilla nanopolish_density_plot.R bc2_window_file.tsv
bc4_window_file.tsv

输出文件:

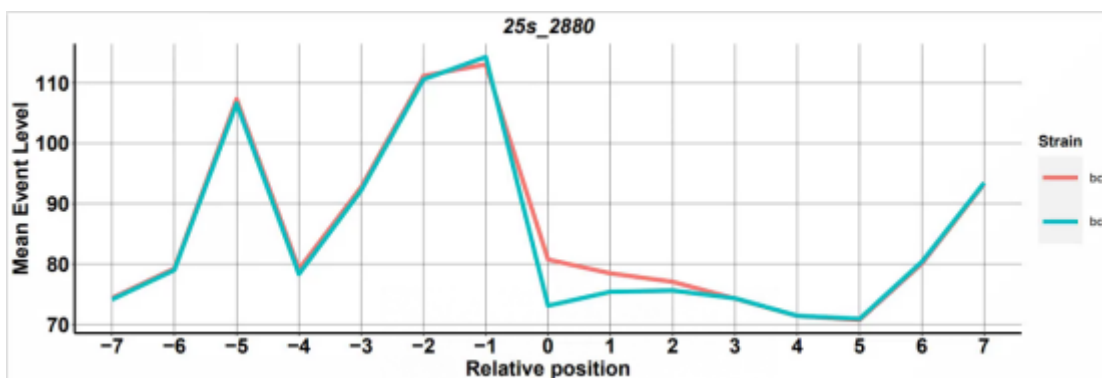


4.2.2. Mean current intensity plots centered in the modified sites

```
Rscript --vanilla nanopolish_meanlineplot.R <window_file1> <window_file2>
<window_file3(optional)> <window_file4(optional)>
```

脚本示例: Rscript --vanilla nanopolish_meanlineplot.R bc2_window_file.tsv
bc4_window_file.tsv

输出文件:

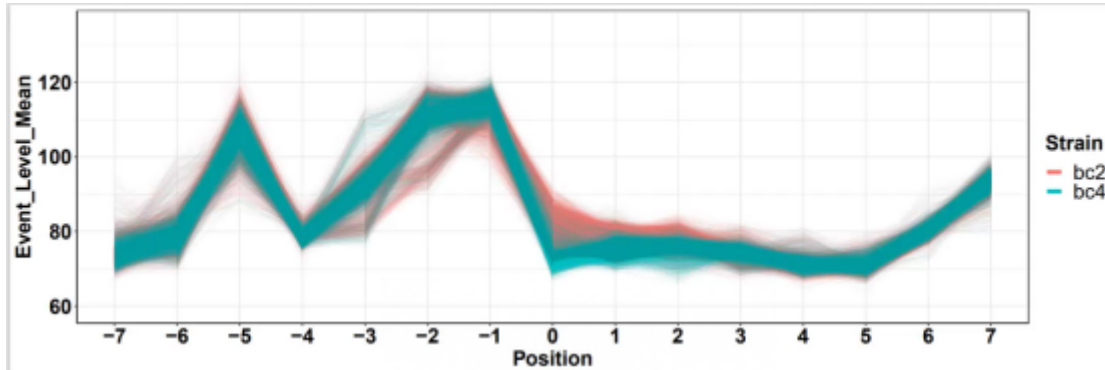


4.2.3 Per-read current intensity plots centered in the modified sites


```
Rscript --vanilla nanopolish_perreadlineplot.R <window_file1> <window_file2>  
<window_file3(optional)> <window_file4(optional)>
```

脚本示例: Rscript --vanilla nanopolish_perreadlineplot.R bc2_window_file.tsv
bc4_window_file.tsv

输出文件:



4.2.4 PCA plots from the per-read 15-mer current intensity data

```
Rscript --vanilla nanopolish_pca.R <window_file1.tsv> <window_file2.tsv>  
<window_file3.tsv(optional)> <window_file4.tsv(optional)>
```

脚本示例: Rscript --vanilla nanopolish_pca.R bc2_window_file.tsv bc4_window_file.tsv

输出文件:

